Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science · Probability for Data Science · Introduction to Urban Data Analytics · *In progress:* Introduction to Machine Learning · Efficient Algorithms & Intractable Problems

Technical Skills

Languages: Python, SQL, Java

ML & Statistics: scikit-learn, XGBoost, LightGBM, PyTorch, Cross-Validation, Feature Engineering

Data: pandas, NumPy, GeoPandas, Matplotlib, seaborn, PostgreSQL, ArcGIS Online, Jupyter

Engineering: FastAPI, REST APIs, AWS S3, pytest, Git, GitHub

Experience

IDX Exchange | Data Science Intern | Remote **Jun 2026 – Aug 2026**

- Developed a model that predicts California home sale prices, typically within **8.6%** of the actual price, using **71K** single-family sales cleaned from **143K** raw records.
- Benchmarked four model types and selected a tuned XGBoost (R^2 of **0.88**), finding that deeper trees improved accuracy more than a lower learning rate.
- Resolved a data leakage bug that had invalidated early results, and nearly halved a simpler model's error (from **29%** to **15%**) by engineering a school district feature.
- Analyzed errors by price band to find weak spots: luxury homes (**13%** error vs. **6.5%** mid-priced) and low-sales towns.

Guang Labs | Engineering Intern | Remote **May 2026 – Aug 2026**

- Engineered a service for an AI video startup that classifies videos and images by type (sports, gaming, talking) using Gemini, retrying uncertain or malformed responses on different clips, up to **10** attempts.
- Persisted each label with the model's confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that alerts users when their video finishes processing or fails.

Fung Institute | Fung Fellow, Health + Innovation | Berkeley, CA **Aug 2026 – Present**

- Selected for Berkeley's industry-partner fellowship, placed on The After Collective project supporting wildfire survivors.
- Interviewed wildfire survivors about the tools and information they use, and organize the team's documentation.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Apify, Factory Droid* GitHub

- Co-designed a transfer planner for California community college students that shows which classes count toward their major using official ASSIST agreements, citing a source for every answer.
- Shaped the product idea, features, and vision, and gave feedback that improved the interface and added more useful features.
- Grounded the app in real data so nothing is made up: credits are matched by code against official records, and campus opportunities are scraped from real pages (Apify, run by a Factory Droid agent) with source links.

Housing Price & Assessment Bias | *Python, pandas, scikit-learn* GitHub

- Built a model that estimates home values from **200K+** Cook County (Chicago) property records.
- Audited its errors across price ranges and found cheaper homes are assessed too high, so their owners are taxed unfairly.

SB 79 Transit Housing Analysis (Los Angeles) | *ArcGIS Online, StoryMaps* GitHub

- Mapped the parts of Los Angeles that qualify for denser housing near transit under a new California law (**6,413** eligible zoning areas) with displacement, pollution, and fire hazard data, finding about **63%** of it is in low-resource areas.

Leadership & Involvement

UC Berkeley CDSS | Reader/Tutor, CDSS 198 | Berkeley, CA **Aug 2026 – Present**

- Give resume feedback in a data science careers seminar, and run exam reviews, practice worksheets, and office hours for transfer students.

UC Berkeley EECS | Course Staff (UCS1), CS 61A | Berkeley, CA **Jun 2026 – Aug 2026**

- Taught **4** weekly exam-prep sections, held office hours, helped write midterm and final questions, and led final review sessions.